

Podcast 30 — IoT, End-User Computing & Niche

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Length target: 15 min **Repo references:** `domain-2-new-solutions/23-aws-iot.md`, `domain-1-organizational/42-workspaces.md`, `domain-2-new-solutions/40-amplify.md`, `46-pinpoint.md`, `47-device-farm.md`, `37-other-services.md` (Connect), `49-niche-services.md`

Topic & Scope

Final cleanup: IoT family (Core / Greengrass / SiteWise / Device Defender / Device Management), end-user computing (WorkSpaces / AppStream / WorkSpaces Web), modernization tooling (Amplify, Pinpoint, Device Farm, Connect), and the niche distractors (Managed Blockchain, Elastic Transcoder).

Service Coverage Depth

Brief — “what it does + when to pick”: - IoT Core (MQTT broker + rules engine) - Greengrass (edge runtime) - SiteWise (industrial telemetry) - Device Defender (IoT security) - Device Management (fleet ops) - WorkSpaces (VDI) - AppStream 2.0 (streamed app) - WorkSpaces Web (browser isolation) - Amplify (full-stack hosting + backend) - Pinpoint (multi-channel marketing engagement) - Device Farm (mobile testing) - Amazon Connect (call center) - Managed Blockchain (distractor) - Elastic Transcoder (legacy distractor)

Structured Outline

1. **Open (30s)** — “Final podcast: IoT, end-user computing, modernization specialties, and the distractor cluster. None of these need deep coverage — but you need to recognize which one fits each scenario.”
2. **IoT Family (4 min)** — IoT Core (MQTT broker + rules engine + device shadow), Greengrass (run Lambda/containers at edge devices), SiteWise (industrial OPC UA / Modbus telemetry + dashboards), Device Defender (audit + detect anomalies), Device Management (fleet provisioning + OTA updates)
3. **End-User Computing (3 min)** — WorkSpaces (persistent VDI), AppStream 2.0 (streamed single app, no full desktop), WorkSpaces Web (browser isolation)
4. **Modernization Specialties (4 min)** — Amplify (full-stack hosting + Cognito/AppSync/DynamoDB backend), Pinpoint (multi-channel marketing campaigns), Device Farm (real mobile devices for testing), Amazon Connect (cloud call center)
5. **Niche Distractors (2.5 min)** — Managed Blockchain (Hyperledger / Ethereum), Elastic Transcoder (legacy — MediaConvert is the modern answer), IoT Things Graph (deprecated)
6. **Rapid-Fire Trap Drill (60s)**

Must-Mention Exam Tips

- IoT Core: MQTT broker, device registry, device shadow (last-known state), rules engine (route to Lambda/S3/Kinesis/DynamoDB)
- IoT Greengrass: run Lambda + containers + ML at the edge, sync with cloud, offline operation
- IoT SiteWise: industrial telemetry, OPC UA / Modbus / EtherNet/IP, asset models, dashboards
- IoT Device Defender: audit IoT policies, detect anomalies (cert sharing, traffic patterns)
- IoT Device Management: fleet provisioning + OTA firmware updates + grouping
- WorkSpaces: persistent desktops (Windows / Linux), needs Directory Service, hourly or monthly billing
- WorkSpaces Pools: ephemeral non-persistent desktops
- AppStream 2.0: streamed single application (e.g., AutoCAD)
- WorkSpaces Web: browser isolation for SaaS access
- Amplify Hosting: CI/CD from Git, PR previews, SSR support, CloudFront-backed
- Amplify Backend (Gen 2): TypeScript schema generates Cognito + AppSync + DynamoDB
- Pinpoint: multi-channel (email + SMS + push + voice + in-app), journeys, segmentation, analytics
- Pinpoint vs SES: Pinpoint is segmented marketing; SES is transactional at scale (cheaper for bulk transactional)
- Device Farm: real iOS / Android devices + desktop browsers, Appium / XCUITest / Selenium
- Amazon Connect: cloud call center, contact flows, integrates with Lex bots
- Managed Blockchain: Hyperledger Fabric + Ethereum networks, multi-party ledger
- Elastic Transcoder: legacy video conversion — for new workloads use MediaConvert
- IoT Things Graph: deprecated — rarely the right answer

Must-Mention Exam Traps

- EXAM TRAP: IoT Core ≠ IoT Greengrass — Core is cloud broker; Greengrass is edge runtime
- EXAM TRAP: SiteWise is INDUSTRIAL (OPC UA, Modbus) — not consumer IoT
- EXAM TRAP: WorkSpaces requires Directory Service (Managed AD, AD Connector, or Simple AD)
- EXAM TRAP: AppStream 2.0 = single app stream, NOT full desktop
- EXAM TRAP: WorkSpaces Web is browser-only, NOT a full desktop
- EXAM TRAP: Amplify is for full-stack apps — Beanstalk for server-based PaaS, App Runner for containers
- EXAM TRAP: Pinpoint ≠ SES — SES is transactional email at scale (cheaper); Pinpoint is multi-channel marketing with analytics
- EXAM TRAP: Pinpoint ≠ SNS — SNS is generic pub/sub; Pinpoint is engagement platform
- EXAM TRAP: Amazon Connect ≠ Pinpoint — Connect is contact center voice/chat; Pinpoint is outbound marketing
- EXAM TRAP: Device Farm ≠ CloudWatch Synthetics — Device Farm is pre-release mobile QA; Synthetics is production probes
- EXAM TRAP: Managed Blockchain is rarely the right answer in SAP-C02 — usually a distractor unless explicit multi-party ledger requirement

- EXAM TRAP: Elastic Transcoder is LEGACY — when MediaConvert is also offered, pick MediaConvert
- EXAM TRAP: IoT Events / IoT Analytics / IoT Things Graph being deprecated or end-of-support — don't pick for new designs
- EXAM TRAP: QLDB is closed to new customers — for append-only audit use DynamoDB with versioning or CloudTrail + S3 Object Lock

Key Decision Matrix

Scenario	Pick
MQTT broker + device shadow + rules	IoT Core
Run Lambda at edge devices, offline-tolerant	IoT Greengrass
Industrial telemetry (OPC UA / Modbus)	IoT SiteWise
IoT security audit + anomaly detection	IoT Device Defender
IoT fleet provisioning + OTA updates	IoT Device Management
Persistent virtual desktop	WorkSpaces
Stream one app to many users	AppStream 2.0
Browser isolation for SaaS access	WorkSpaces Web
Full-stack web/mobile app with CI/CD + backend	Amplify
Marketing campaign multi-channel + segmentation	Pinpoint
Bulk transactional email at scale	SES (NOT Pinpoint)
Cloud call center with voice + chat	Amazon Connect
Test mobile app on real devices	Device Farm
Multi-party Hyperledger / Ethereum	Managed Blockchain
Modern video transcoding	MediaConvert (NOT Elastic Transcoder)

Tone & Style

- Each service gets a one-sentence definition + one use case
- Frame the distractor section explicitly: “These show up as wrong answers — recognize and skip”
- Repeat “Pinpoint vs SES” since it’s a recurring confusion

Rapid-Fire Closer

“MQTT broker?” — “IoT Core.” “Edge Lambda?” — “Greengrass.” “Stream one app?” — “AppStream.” “Marketing campaigns?” — “Pinpoint.” “Bulk transactional email?” — “SES.” “Call center?” — “Connect.” “Hyperledger?” — “Managed Blockchain.” “Mobile device testing?” — “Device Farm.”